Griffiths Chapter 10: Gauge Theories

A First-Exposure Summary of Sections 10.5–10.6

PHYS 7502 Reading Guide

1 Section 10.5 — Chromodynamics

Where We Are

In Section 10.3, we demanded local $U(1)$ invariance $\to$ got QED (1 gauge field = photon).
In Section 10.4, we demanded local $SU(2)$ invariance $\to$ got Yang–Mills theory (3 gauge fields).
Now: demand local $SU(3)$ invariance $\to$ QCD (8 gauge fields = gluons).

The Setup: Three Colors of Quarks

Each quark flavor (up, down, strange, ...) comes in three **colors**: red, blue, green. For a single flavor, the free Lagrangian is:

$$L = i\hbar c\,\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - mc^2\bar{\psi}\psi \tag{10.72}$$

where ψ is now a **three-component column vector** in color space:

$$\psi = \begin{pmatrix} \psi_r \\ \psi_b \\ \psi_g \end{pmatrix} \tag{10.71}$$

Term Refresher: Color

"Color" has absolutely nothing to do with visible color. It is a label for the three-valued charge of the strong interaction. Each entry ψ_r, ψ_b, ψ_g is itself a full four-component Dirac spinor, so ψ is really a 12-component object (3 colors $\times$ 4 spinor components).

The Symmetry Group: $U(3) = U(1) \otimes SU(3)$

The Lagrangian (10.72) is invariant under $\psi \to U\psi$ for any constant unitary 3×3 matrix U. Any such matrix can be decomposed as:

$$U = e^{i\theta} \cdot e^{i\boldsymbol{\lambda}\cdot\mathbf{a}/2}$$

- The $e^{i\theta}$ part is the $U(1)$ phase we already handled in QED.
- The $e^{i\boldsymbol{\lambda}\cdot\mathbf{a}/2}$ part, with $\det = 1$, belongs to $SU(3)$. This is the *new* symmetry.

Term Refresher: Gell-Mann Matrices

The **Gell-Mann matrices** $\lambda_1, \lambda_2, \ldots, \lambda_8$ are to $SU(3)$ what the Pauli matrices are to $SU(2)$. They are eight 3×3, traceless, Hermitian matrices that serve as the **generators** of $SU(3)$.

Recall the analogy:

	$SU(2)$	$SU(3)$
Generators	τ_1, τ_2, τ_3 (Pauli)	$\lambda_1, \ldots, \lambda_8$ (Gell-Mann)
Size	2×2	3×3
Number	3	8
Commutator	$[\tau_a, \tau_b] = 2i\epsilon_{abc}\tau_c$	$[\lambda_a, \lambda_b] = 2if_{abc}\lambda_c$

The f_{abc} are the **structure constants** of $SU(3)$. They are completely antisymmetric, and most of them are zero. The nonzero ones include $f_{123} = 1$, $f_{147} = f_{246} = f_{257} = f_{345} = f_{516} = f_{637} = \frac{1}{2}$, and $f_{458} = f_{678} = \frac{\sqrt{3}}{2}$.

Making It Local

We now demand that the Lagrangian be invariant under *local* $SU(3)$ transformations:

$$\psi \;\to\; S(x)\,\psi, \qquad S(x) = e^{-iq\boldsymbol{\lambda}\cdot\boldsymbol{\phi}(x)/\hbar c} \tag{10.79}$$

The procedure is identical in spirit to what we did for $U(1)$ and $SU(2)$:

1. Replace ∂_μ with the covariant derivative:

$$D_\mu = \partial_\mu + \frac{iq}{\hbar c}\,\boldsymbol{\lambda}\cdot\mathbf{A}_\mu \tag{10.80}$$

   where $\mathbf{A}_\mu = (A^1_\mu, A^2_\mu, \ldots, A^8_\mu)$ are **eight** gauge fields.
2. Assign a transformation rule to $\mathbf{A}_\mu$ so that $D_\mu\psi$ transforms the same way as ψ. For infinitesimal transformations:

$$A^i_\mu \;\to\; A^i_\mu + \partial_\mu\phi^i + \frac{2q}{\hbar c}\sum_{j,k} f_{ijk}\,\phi_j\,A^k_\mu \tag{10.83}$$

   This is the same structure as Yang–Mills (Eq. 10.61), but with ϵ_{ijk} replaced by the $SU(3)$ structure constants f_{ijk}.
3. Add the free Lagrangian for the gauge fields.

The Field Strength Tensor

Just as in Yang–Mills theory, the field strength tensor picks up a nonlinear term:

$$\mathbf{F}_{\mu\nu} = \partial_\mu\mathbf{A}_\nu - \partial_\nu\mathbf{A}_\mu - \frac{2q}{\hbar c}\,(\mathbf{A}_\mu\times\mathbf{A}_\nu) \tag{10.87}$$

where the "cross product" is now defined using the $SU(3)$ structure constants:

$$(\mathbf{B}\times\mathbf{C})_i = \sum_{j,k=1}^{8} f_{ijk}\,B_j\,C_k \tag{10.84}$$

Don't confuse with ordinary cross product

This is *not* the cross product from vector calculus (which only works in 3D). The indices run from 1 to 8 (over gluon types), and the structure constants f_{ijk} replace the Levi-Civita symbol ϵ_{ijk}. The notation $\times$ is Griffiths' shorthand, and it works the same way for $SU(2)$ (where $f_{ijk} = \epsilon_{ijk}$) and $SU(3)$ (where f_{ijk} is more complicated).

The Complete QCD Lagrangian

Chromodynamics Lagrangian

$$\boxed{L = \left[i\hbar c\,\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - mc^{2}\bar{\psi}\psi\right] - \frac{1}{16\pi}\mathbf{F}^{\mu\nu}\cdot\mathbf{F}_{\mu\nu} - q\left(\bar{\psi}\gamma^{\mu}\boldsymbol{\lambda}\psi\right)\cdot\mathbf{A}_{\mu}} \tag{10.88}$$

This describes three equal-mass Dirac fields (the three colors of one quark flavor) interacting with eight massless vector gauge fields (the gluons). Six replicas of this Lagrangian (one per flavor, each with the appropriate quark mass) give the full QCD Lagrangian.

Comparing the three gauge theories

	QED	**Yang–Mills**	**QCD**
Symmetry group	$U(1)$	$SU(2)$	$SU(3)$
Generators	1	3 (Pauli)	8 (Gell-Mann)
Gauge fields	1 (photon)	3	8 (gluons)
Matter fields	1 Dirac	2 Dirac	3 Dirac (colors)
Self-interaction?	No	Yes	Yes
Abelian?	Yes	No	No

Key Physical Consequences

Gluons carry color charge

Because $SU(3)$ is non-Abelian, the field strength $\mathbf{F}_{\mu\nu}$ contains the cross-product term $\mathbf{A}_{\mu}\times\mathbf{A}_{\nu}$. This means the "free" gauge Lagrangian $\mathbf{F}^{\mu\nu}\cdot\mathbf{F}_{\mu\nu}$ already contains cubic and quartic terms in $\mathbf{A}_{\mu}$, giving rise to:



3-gluon vertex 4-gluon vertex

Photons don't interact with each other (at tree level). Gluons *do* — they carry color charge, so they are themselves sources of the color field. This is the root cause of **confinement** and **asymptotic freedom**.

Why only 8 gluons

You might expect $3 \times 3 = 9$ color–anticolor combinations. But the color-singlet combination $\frac{1}{\sqrt{3}}(r\bar{r} + b\bar{b} + g\bar{g})$ does not couple to color charge — it would mediate a long-range strong force between colorless hadrons, which is not observed. So QCD has $SU(3)$ symmetry (8 generators), not $U(3)$ symmetry (9 generators).

Color Currents

The Dirac fields generate eight color currents (one per gluon type):

$$J_a^\mu = cq\,\bar{\psi}\gamma^\mu\lambda_a\psi \qquad (a = 1, 2, \ldots, 8) \tag{10.89}$$

These are the QCD analogue of the electromagnetic current $J^\mu = cq\,\bar{\psi}\gamma^\mu\psi$. The λ_a matrices shuffle the color components of ψ, so the currents describe quarks changing color as they emit/absorb gluons.

2 Section 10.6 — Feynman Rules from Lagrangians

The Question

We've been writing down Lagrangians. How do we get from a Lagrangian to the Feynman rules we've been using to calculate cross sections?

The Big Idea

The passage from classical field theory to quantum field theory involves **quantizing** the fields — reinterpreting the field variables so that particles emerge as quanta of the fields. The quantization procedure is technically involved (second quantization, canonical commutation relations, etc.), but the *end result* is simple:

Free Lagrangian $\Longrightarrow$ Propagator
Interaction terms $\Longrightarrow$ Vertex factors

That is: you can *read off* the Feynman rules directly from the Lagrangian, without going through the full quantization.

Step 1: Propagators from Free Lagrangians

The free Lagrangians give the free field equations (via Euler–Lagrange):

$$\left[\partial_\mu\partial^\mu + \left(\frac{mc}{\hbar}\right)^2\right]\phi = 0 \qquad \text{Klein–Gordon (spin 0)} \qquad (10.13)$$

$$\left[i\gamma^\mu\partial_\mu - \frac{mc}{\hbar}\right]\psi = 0 \qquad \text{Dirac (spin } \tfrac{1}{2}\text{)} \qquad (10.15)$$

$$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) + \left(\frac{mc}{\hbar}\right)^2 A^\nu = 0 \qquad \text{Proca (spin 1)} \qquad (10.22)$$

Going to Momentum Space

The standard prescription $\partial_\mu \leftrightarrow -\frac{i}{\hbar}p_\mu$ converts the differential equations above into algebraic equations. In momentum space, the field equations become:

$$\left[p^2 - (mc)^2\right]\phi = 0 \qquad (10.90)$$

$$\left[p - mc\right]\psi = 0 \qquad (10.91)$$

$$\left[-(p^2 - (mc)^2)g^{\mu\nu} + p^\mu p^\nu\right]A_\nu = 0 \qquad (10.92)$$

where $p \equiv \gamma^\mu p_\mu$ is the Feynman slash.

The Rule: Propagator

The propagator for each spin is obtained by taking the factor in square brackets from the momentum-space field equation and computing its inverse, multiplied by i:

The Three Propagators

Spin-0 Propagator (Klein–Gordon)

$$\boxed{\frac{i}{p^2-(mc)^2}} \tag{10.93}$$

The bracket was the scalar $[p^2-(mc)^2]$. Its "inverse" is just $1/[p^2-(mc)^2]$.

Spin-$\frac{1}{2}$ Propagator (Dirac)

$$\boxed{\frac{i(p+mc)}{p^2-(mc)^2}} \tag{10.94}$$

The bracket was the 4×4 matrix $[p-mc]$. Its matrix inverse is $(p+mc)/[p^2-(mc)^2]$, since $(p-mc)(p+mc)=p^2-(mc)^2$.

Spin-1 Propagator (Proca

$$\boxed{\frac{-i}{p^2-(mc)^2}\left[g^{\mu\nu}-\frac{p^\mu p^\nu}{(mc)^2}\right]} \tag{10.95}$$

The bracket was a second-rank tensor $T^{\mu\nu}$; you need its tensor inverse $(T^{-1})_{\mu\nu}$ such that $T^{\mu\alpha}(T^{-1})_{\alpha\nu}=\delta^\mu{}_\nu$.

The photon propagator is special

You *cannot* set $m\to 0$ in the Proca propagator — it blows up because of the $p^\mu p^\nu/(mc)^2$ term. Instead, go back to the massless field equation $\partial_\mu F^{\mu\nu}=0$ and impose the **Lorentz gauge** $\partial_\mu A^\mu=0$, which reduces the equation to $\partial^2 A^\nu=0$, giving:

$$\text{Massless spin-1 propagator:}\quad \boxed{\frac{-ig^{\mu\nu}}{p^2}} \tag{10.99}$$

The $g^{\mu\nu}$ (rather than a more complicated tensor) is a consequence of the specific gauge choice.

Physical meaning of the propagator

The propagator $\sim i/(p^2-m^2)$ describes a **virtual particle** carrying four-momentum p. For a real (on-shell) particle, $p^2=m^2$ and the propagator diverges — reflecting the fact that a real particle propagates forever, not just between two vertices. Virtual particles have $p^2\neq m^2$ (they are "off shell"), and the propagator tells you how the amplitude falls off as the particle gets further from its mass shell.

Step 2: Vertex Factors from Interaction Terms

The Recipe

1. Identify the interaction terms L_{int} in the Lagrangian (everything beyond the free-field terms).
2. Write iL_{int} in momentum space ($\partial_\mu \to -ip_\mu/\hbar$).
3. Look at which fields appear — this tells you which lines meet at the vertex (one line per field variable).
4. "Rub out" the field variables. What's left is the **vertex factor**.

Example: QED Vertex

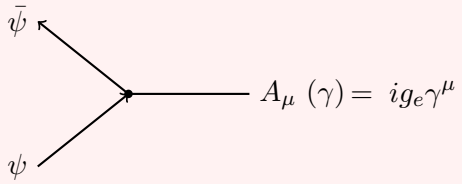
The QED interaction term [from the Lagrangian (10.35)] is:

$$L_{\text{int}} = -q\,\bar{\psi}\gamma^\mu\psi\,A_\mu$$

Three fields are present: $\bar{\psi}$ (outgoing fermion), ψ (incoming fermion), and A_μ (photon). So the vertex joins three lines.

Multiply by i, rub out the fields, and account for the cgs photon normalization factor:

QED Vertex Factor



where $g_e = \sqrt{4\pi\alpha\hbar c}$ for electrons (with $q = -e$). This is the vertex factor we've been using since Chapter 7.

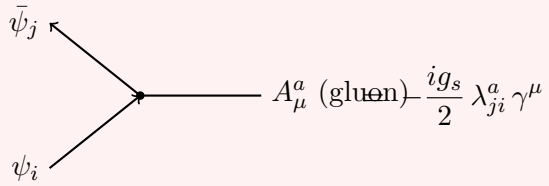
Example: QCD Quark–Gluon Vertex

From the QCD Lagrangian (10.88), the quark–gluon interaction term is:

$$L_{\text{int}} = -q\,\bar{\psi}\gamma^\mu\boldsymbol{\lambda}\psi\cdot\mathbf{A}_\mu$$

Same structure as QED, but with the Gell-Mann matrix λ_a sandwiched between the spinors, and the gluon field A^a_μ carrying a color index a.

QCD Vertex Factor



where $g_s = 2\sqrt{4\pi q/\hbar c}$ is the strong coupling constant, i, j are color indices, and a labels the gluon type (1–8).

The λ^a_{ji} factor is just a number

For a given color combination ($i \to j$ via gluon a), the Gell-Mann matrix element λ^a_{ji} is simply a number (an entry from one of the eight 3×3 matrices). The Dirac structure γ^μ is the same as in QED. So QCD vertices look like QED vertices, but each one carries an extra numerical "color factor."

Example: Gluon Self-Interaction Vertices

The field strength tensor $\mathbf{F}_{\mu\nu}$ contains not only the "free" part $\partial_\mu \mathbf{A}_\nu - \partial_\nu \mathbf{A}_\mu$ but also the interaction term $-\frac{2q}{\hbar c}(\mathbf{A}_\mu \times \mathbf{A}_\nu)$. When we form $\mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu}$, squaring out produces:

$$L_{\text{int}}^{(\text{gluon})} = \underbrace{\frac{q}{4\pi\hbar c}\left[(\partial_\mu \mathbf{A}_\nu - \partial_\nu \mathbf{A}_\mu) \cdot (\mathbf{A}^\mu \times \mathbf{A}^\nu)\right]}_{\text{3 factors of } A \to \text{3-gluon vertex}} - \underbrace{\frac{q^2}{4\pi(\hbar c)^2}(\mathbf{A}_\mu \times \mathbf{A}_\nu) \cdot (\mathbf{A}^\mu \times \mathbf{A}^\nu)}_{\text{4 factors of } A \to \text{4-gluon vertex}} \tag{10.104}$$

No analogue in QED!

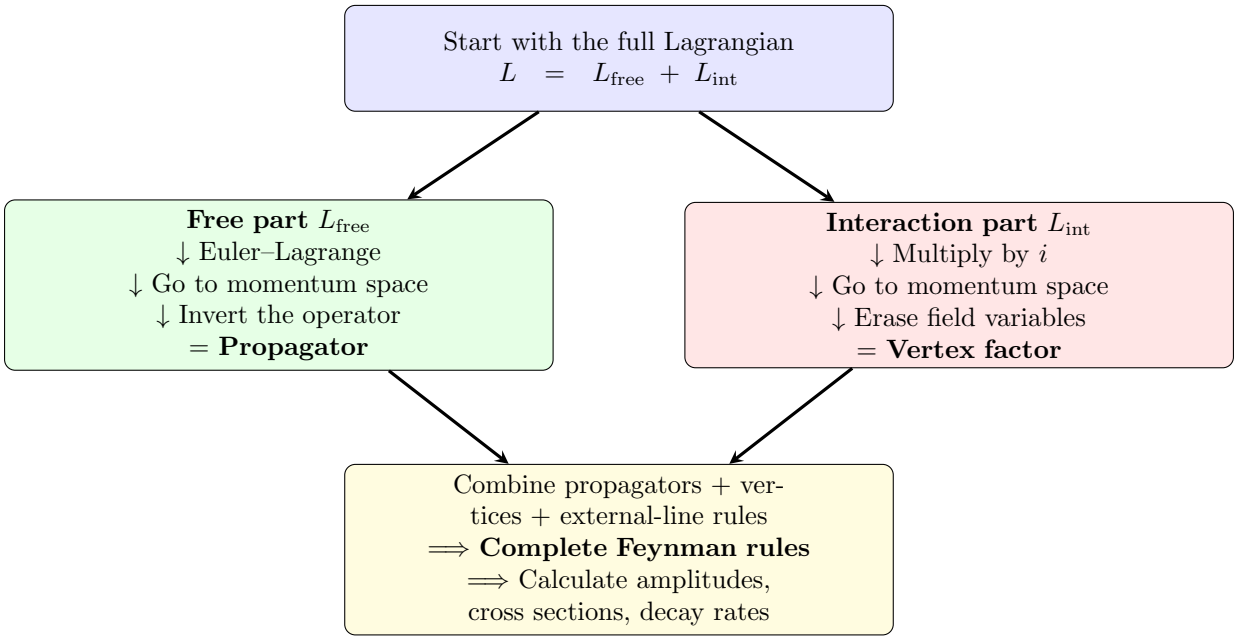
These self-interaction vertices exist *only* because $SU(3)$ is non-Abelian. In QED, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is *linear* in A, so $F^{\mu\nu}F_{\mu\nu}$ is quadratic (free field, no interactions). In QCD (and Yang–Mills), the nonlinear piece $\mathbf{A} \times \mathbf{A}$ in $\mathbf{F}_{\mu\nu}$ generates cubic and quartic interaction terms.

Physically: photons don't talk to each other, but gluons do. This is the origin of asymptotic freedom and confinement.

Complete Feynman Rules: Summary Table

Element	**From...**	**Rule**
Spin-0 propagator	Free Klein–Gordon	$\dfrac{i}{p^2 - (mc)^2}$
Spin-$\frac{1}{2}$ propagator	Free Dirac	$\dfrac{i(p + mc)}{p^2 - (mc)^2}$
Massive spin-1 propagator	Free Proca	$\dfrac{-i}{p^2 - (mc)^2}\left[g_{\mu\nu} - \dfrac{p_\mu p_\nu}{(mc)^2}\right]$
Photon propagator	Free Maxwell + gauge	$\dfrac{-ig_{\mu\nu}}{p^2}$
QED vertex	$L_{\text{int}} = -q\bar{\psi}\gamma^\mu\psi A_\mu$	$ig_e\gamma^\mu$
QCD qqg vertex	$L_{\text{int}} = -q\bar{\psi}\gamma^\mu \boldsymbol{\lambda}\psi \cdot \mathbf{A}_\mu$	$-\frac{ig_s}{2}\lambda^a_{ji}\gamma^\mu$
3-gluon vertex	$\mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu}$ (cubic in A)	(see Eq. 8.42)
4-gluon vertex	$\mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu}$ (quartic in A)	(see Eq. 8.43)

Master Summary: The Logic of Section 10.6



The Big Takeaway

The Lagrangian is the *master object* of quantum field theory. Once you have it, the Feynman rules — and hence all observable physics — follow by a systematic recipe. Sections 10.1–10.4 showed how *symmetry* dictates the Lagrangian; Section 10.6 shows how the Lagrangian dictates the *rules of calculation*. Together, they close the loop:

Symmetry $\xrightarrow{\text{gauge invariance}}$ **Lagrangian** $\xrightarrow{\text{Sec. 10.6 recipe}}$ **Feynman Rules** $\xrightarrow{\text{Chapters 6–9}}$ **Cross Sections**